

MUTHYA AMBATI

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WORK EXPERIENCE

Microchip Technology Inc

San Jose, CA

Senior Validation Engineer - I

June 2024 – Present

- Developed RTL for high-performance soft-IPs, including a Fabric AXI4 master and a custom one-logic-level embedded logic analyzer; supported verification and validation in the lab, performed STA, and achieved a highly pipelined design meeting the 500MHz target frequency on PolarFire2X FPGA.
- Validated Ethernet links at 25G, 50G and 100G on the MAC/PCS datapath, by developing tests for link establishment through lane mapping, swizzle logic, AN/LT; and verified error handling and recovery by implementing RS-FEC error handling tests and soak tests for cable pull-and-restore, with the help of an Ethernet Traffic generator/analyzer.
- Brought up MACsec network-encryption validation use cases for AES bulk encryption. Defined ECC error-injection tests on the encrypt and decrypt SA RAMs to verify error handling in the security datapath for ethernet traffic.
- Automated Ethernet soak testing on Viavi test equipment by scripting traffic configurations, developed an Ethernet Validation Assistant chatbot and built an AI-assisted Python tooling to streamline Jira-to-Excel project tracking, boosting team efficiency and data management by using AI tools.

Advance Micro Devices (AMD-Xilinx)

San Jose, CA

Product Development Intern

Sept 2023 – Dec 2023

- Developed an optimised Stamp and Repeat wrapper module, using OOP concepts to manage location constraints, to automate placement and routing of kernels on the 50x8 Versal FPGA AI Engine array for multi-layered ML architectures, reducing compilation time by 10% (*approx*) and improving scalability for high-compute applications.

Secure and Advanced Computer Architecture Lab

Raleigh, NC

Graduate Research Assistant

Jan 2023 – Apr 2023

- Contributed to the Structural Simulation Toolkit (SST), an HPC simulator, by developing Python scripts to auto-generate RTL-based SST components and integrating multiple RTL/C model instances for co-simulation, then analysed performance metrics including IPC and memory transactions across varying memory configurations.

Gilbarco Veeder Root

Mumbai, India

Embedded Engineer

Aug 2020 – July 2022

- Developed and debugged STM32 (32-bit Arm Cortex-M) firmware including custom GPIO, RS485, USART, and I2C drivers optimised to minimise field EMI/EMC noise and designed an isolated, optocoupler-based RS485 circuit in Altium Designer, validating protocol compliance and signal integrity post-fabrication via DSO oscilloscope.

ACADEMIC PROJECTS

RTL Design for Deep Neural Network | System Verilog

- Implemented a synthesizable multi-stage neural network hardware design, interfaced with SRAM for data transfer. Achieved an area-delay product of 270E6 in 1042 clock cycles at an 11ns clock period through timing analysis.

Out-of-Order Superscalar Process Simulator | C++

- Devised a simulator for a 9-stage superscalar processor including register renaming that fetches N instructions per cycle; evaluated the instructions-per-cycle (IPC) trend by varying issue queue and re-order buffer size across different traces.

Coherence Protocols-based Cache Simulator | C++

- Designed a cache simulator for different bus-based coherence protocols like MSI, MESI, and MESI with a snoop filter, and studied various metrics such as the number of bus signals and memory transactions.

Cache Hierarchy Simulator | C++

- Designed a cache hierarchy simulator based on WBWA write policy and LRU replacement policy, achieving the lowest average access time of 0.71 ns, leading to an increase in total area by 157% while implementing both L1, L2 cache levels.

TECHNICAL SKILLS

- Programming Skills & Technologies:** System Verilog, C++, Python, C, TCL
- Protocols:** AXI4, AXI-Stream, SerDes, Ethernet, I2C, SPI, UART
- Relevant Skills:** RTL Design, ASIC/FPGA Flows, Computer Architecture, Cache Coherence Protocols

EDUCATION

North Carolina State University

Raleigh, North Carolina

Masters in Computer Engineering, GPA: 3.9/4.0

Aug 2022 – May 2024

Visvesvaraya National Institute of Technology

Nagpur, India

Bachelor of Technology, Electronics and Communication Engineering, GPA: 8.41/10

Aug 2016 – May 2020

PATENTS AND PUBLICATIONS

- “Dynamic Trigger Selection and Routing in an FPGA”, U.S. Patent Application No. 19/557,429 (filed Mar 2026), Co-inventor – Patent Pending.
- “Outlier Detection Using One-Class Classification” Muthya Ambati, Sonali Gupta, Srikanth Boddu, *Applications of Advanced Computing in Systems*, Springer, 2021.